

Dssr_select is a PyMOL plugin that is able to bridge DSSR (Dissecting the Spatial Structure of RNA) with PyMOL's environment and visualization capabilities. It gives users many unique and novel features:

1. Selecting structural RNA/DNA features: On a loaded structure in PyMOL, users are able to select from up to 17 supported features including things like base pairs, hairpins, stems, junctions, u-turns. G-quadruplexes, pseudoknots, and more such that they are highlighted by PyMOL in its viewer system.
2. Create cartoon block schematics of nucleic acid structures
3. Extract information for sequences on any loaded structure
4. Interact with a 2D secondary structure diagram that mirrors selections with the 3D PyMOL viewer

In order to use Dssr_select, the following prerequisites must be met:

- ☐ PyMOL (with Qt GUI, a standard in the vast majority of modern PyMOL versions)
- ☐ x3dna-dssr DSSR (Dissecting the Spatial Structure of RNA) executable installed such that it is accessible the system PATH and reachable

Dssr_select operates with four key commands:

1. Dssr_select: allows users to select a structural feature of RNA. It is the core command of Dssr_select that uses DSSR to analyze the loaded structure, find the given feature and index of that feature provided by a user and generate a custom PyMOL selection that highlights the feature. For a full explanation see dssr_select command section.
2. Dssr_gui: launches the graphical interface. This command opens the GUI window that allows users to select features with a variety of buttons and widgets in its interface. It can also be accessed from the PyMOL plugin menu. For more information on dssr_gui, go to the dssr_gui section.
3. Dssr_block: Creates nucleic acid block-cartoon schematics. The command is run to create rectangular representations of the bases and display their orientation, planarity, edges, and more to the user. See the dssr_block section for more information.
4. dssr_2d: opens the 2D secondary structure editor. It essentially opens the GUI for the 2D RNA secondary structure panel without using the dssr_gui command or going through the PyMOL plugin section. For more information, see the dssr_2d section.

To install and successfully use Dssr_select, the following prerequisites must be met:

1. PyMOL must be downloaded and available (a modern version so that it has PyQt5 and will be compatible with the Dssr_select script).
 - a. Any modern open source version should work
2. DSSR must be downloaded and have its path available to the user
 - a. Can be acquired from Columbia Technology Ventures (CTE): [DSSR: an integrated software tool for dissecting the spatial structure of RNA](#)

Once the prerequisites for Dssr_select are met, the user must download the dssr_select.py Python script from its public github repository: [xiang-jun/dssr-pymol: Official PyMOL plugin for DSSR: Interactive selection and 3D visualization of RNA/DNA structural motifs \(stems, pseudoknot layers, hairpins, and more\) with an integrated Qt GUI](https://github.com/xiang-jun/dssr-pymol). After the Python script is downloaded, the user must open PyMOL and navigate to the Plugin Manager where a new plugin can be installed and the dssr_select.py script can be installed. After doing this, Dssr_select should be functional for users.

Using Dssr_select from the Command Line (all commands):

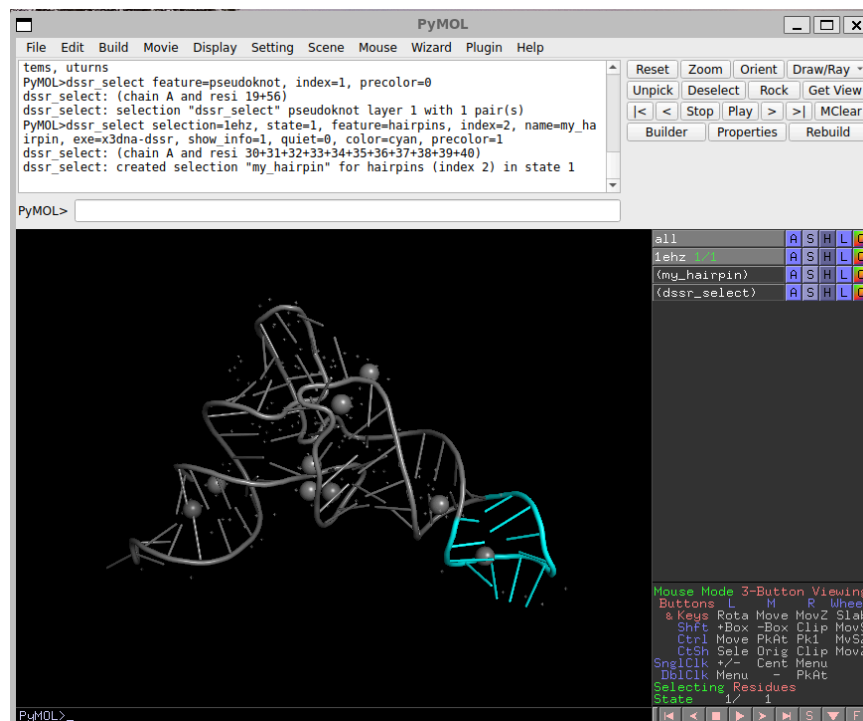
- dssr_select: the command 'dssr_select' is used to select a feature from the already loaded nucleic acid structure in PyMOL. The command takes a total of 10 parameters:

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Parameter	Default	Meaning
selection	"all"	Which atoms in PyMOL to analyze
state	-1	What state to use (-1 means current)
feature	"pairs"	The type of nucleic acid feature/element to select
index	1	The instance of the feature (1 means the first instance)
name	"dssr_select"	The name of the selection object that appears when the command is called
exe	"x3dna-dssr"	The path to the DSSR executable
show_info	0	Whether or not to print extra DSSR output to the console (0 tells it not to print)
quiet	1	Controls if the command will print a message to the console (1 means no

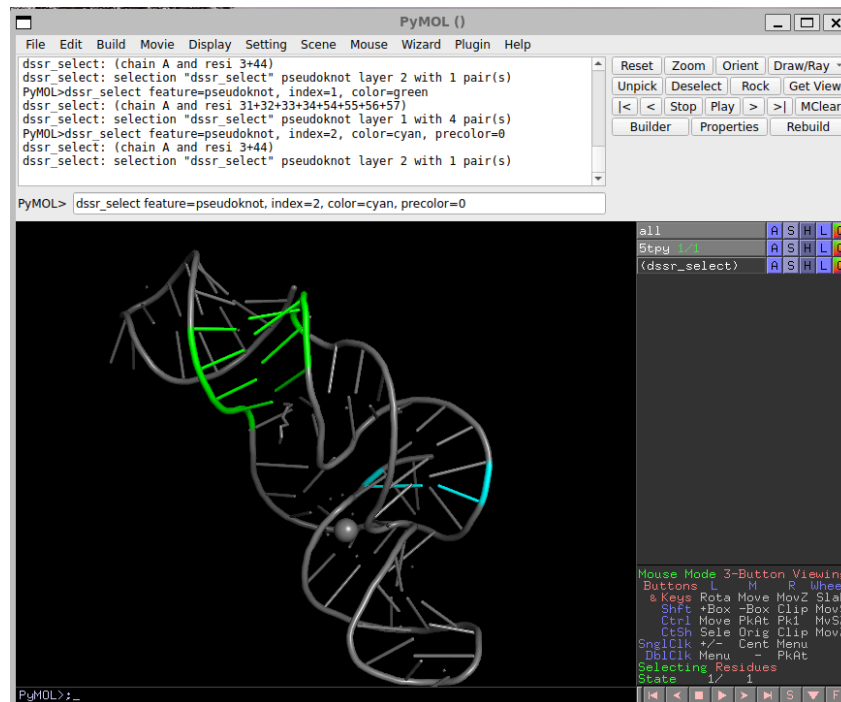
		output)
color	"auto"	The color that is applied to the selected features. ("auto" means most features by default will get pink, pseudoknot layers get a certain color based on each layer)
precolor	1	When set to 1, the entire structure is set to gray before a feature is highlighted to show contrast. When set to 0, the structure stays the same.

- To use all possible parameters, run the following commands after loading in a feature:
 - fetch 1ehz
 - dssr_select selection=1ehz, state=1, feature=hairpins, index=2, name=my_hairpin, exe=x3dna-dssr, show_info=1, quiet=0, color=cyan, precolor=1

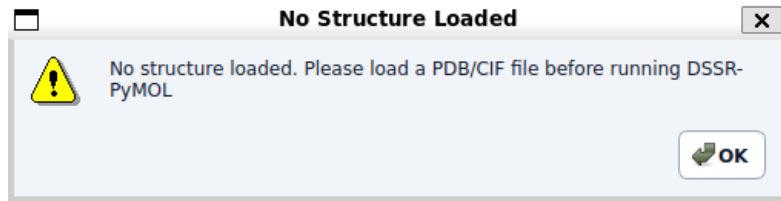


- These parameters allow the user to control numerous parts of the selection that they want to create. One notable way to modify the parameters is to select multiple features. For example, if a user wanted to see each layer of a pseudoknot in a different color, they could run the following commands:

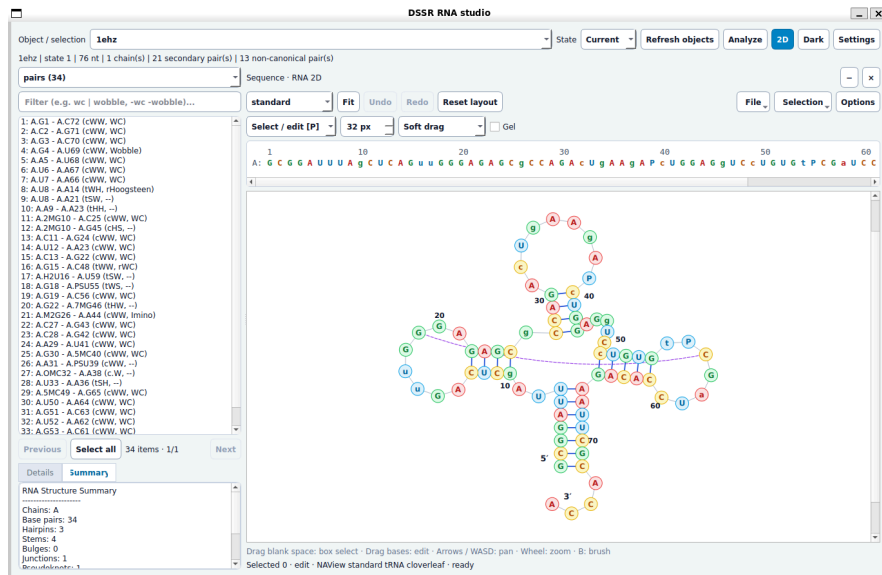
- fetch 5tpy
- dssr_select feature=pseudoknot, index=1, color=green
- dssr_select feature=pseudoknot, index=2, color=cyan, precolor=0



- By using precolor in this instance, the user is able to save the first selection they made that highlights the first layer of the pseudoknot(s) in the fetched structure, 5tpy. As a result, they are able to view both the first and second layers of pseudoknot(s) in 5tpy because they used precolor=0 to prevent dssr_select from overriding the first selection with the second and saving its coloring.
 - Using dssr_select from the command line of PyMOL can be a very effective way of selecting certain features in structures. However, that is not to say that it is perfect. During the designing stage of dssr_select, it was determined that a GUI (Graphical User Interface) to be paired with the command line would be effective at making the tool as useful as possible. This is why dssr_2d/dssr_gui was developed.
- dssr_gui/dssr_2d: the 'dssr_gui' and 'dssr_2d' command launches the Graphical User Interface (GUI) of dssr_select to simplify the process of selecting features and make it more customized to the user.
 - To launch the GUI of dssr_select, first ensure that a structure is loaded in. If a user tries to activate the GUI without a PyMOL structure loaded first, they will be met with the following message:

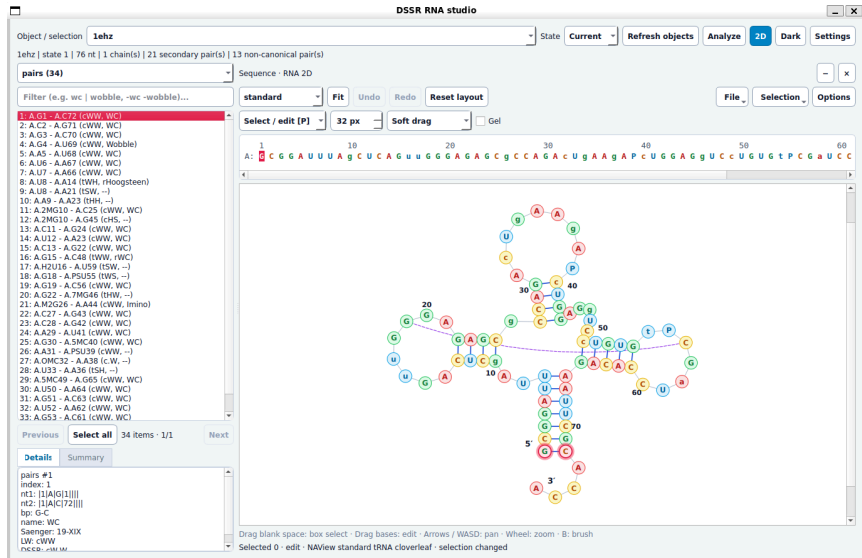


- One way to activate the GUI is to run the command: 'dssr_gui' or 'dssr_2d' in the PyMOL command line. Another way is to click on the tab labeled "Plugin" in PyMOL's menu and click on the button labeled "DSSR" which will then activate the GUI. Nonetheless, once a feature is loaded (take 1ehz for example), and the GUI is activated, the following window will appear (separate from the PyMOL viewer):

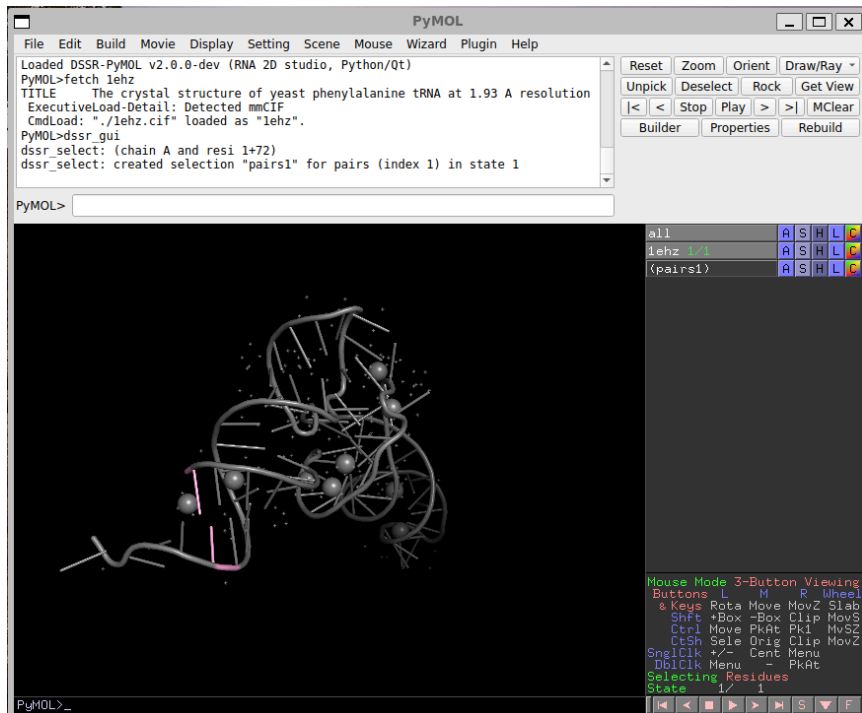


- Selecting features is made simple and efficient using the drop-down menu on the left-hand side of the page. In this example, "pairs (34)" is shown. This denotes that there are 34 indices of pairs to which the user can select all shown in the list below. Selecting one of the indices of features by clicking on them in the labeled list will highlight the feature on both the 2D diagram shown in the right-hand side of the screen and on PyMOL's 3D viewer.

■ Sample: Click index 1: A.G1 - A.C72 (cWW, WC):

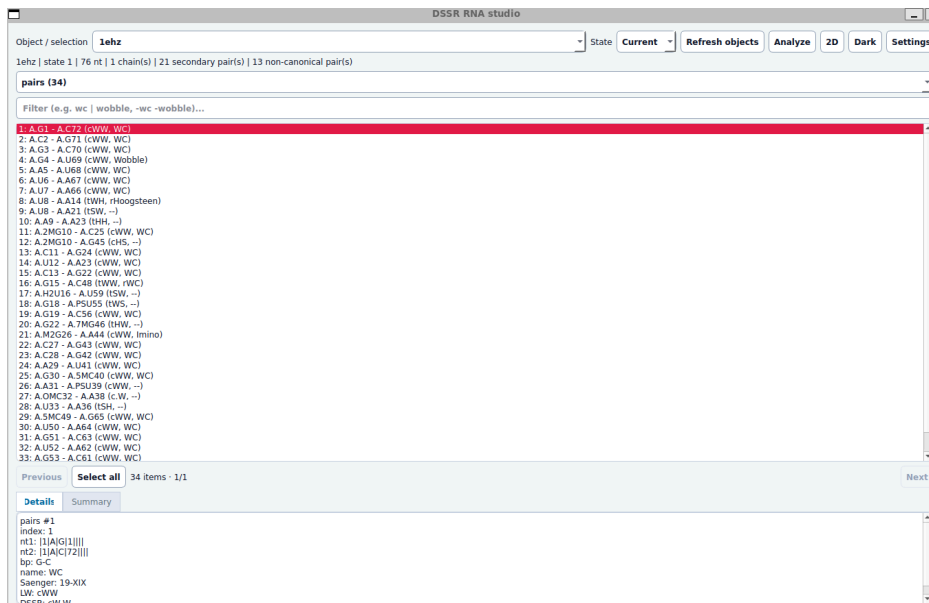


- In this screenshot, it is clear which parts of the diagram have been selected by the user since the first index of a pair in 1ehz is highlighted in red both in the list of indices and on the 2D diagram showing G-C

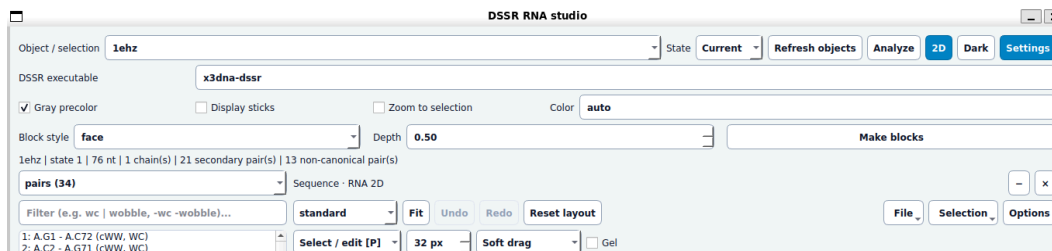


- In this screenshot, it is clear that the first index of pair in 1ehz is highlighted in pink due to the user's selection from the GUI. In essence, selections from the GUI show up both in the 2D diagram and in PyMOL's 3D molecular viewer.
- To use the GUI without the 2D diagram of the selected structure, de-select the 2D button in the top right corner which is selected by default when the GUI is opened. With respect to selecting features and showing them in PyMOL's 3D

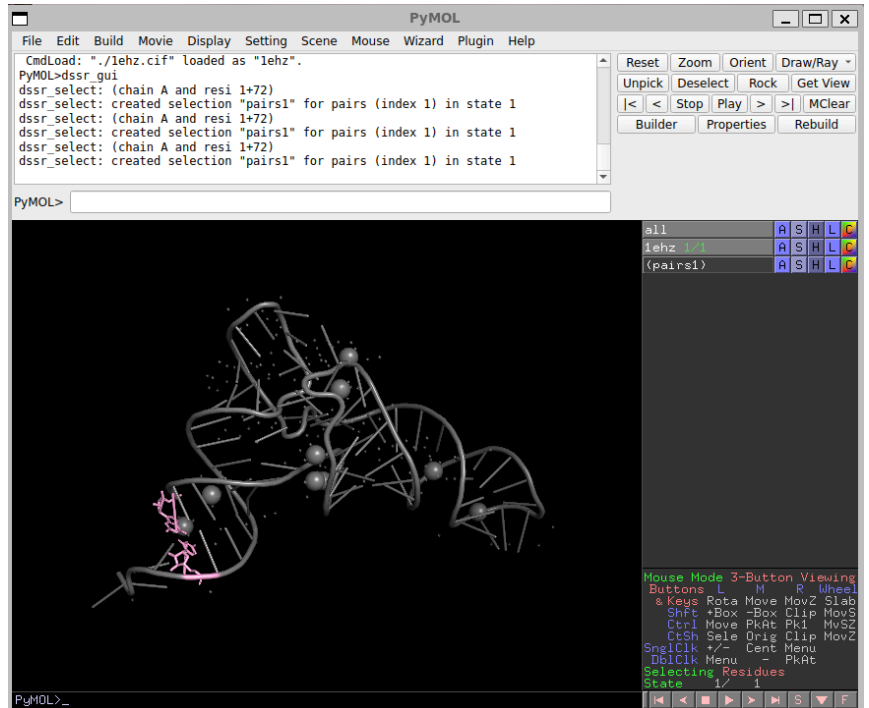
viewer, nothing will change, but the 2D diagram will disappear unless toggled back.



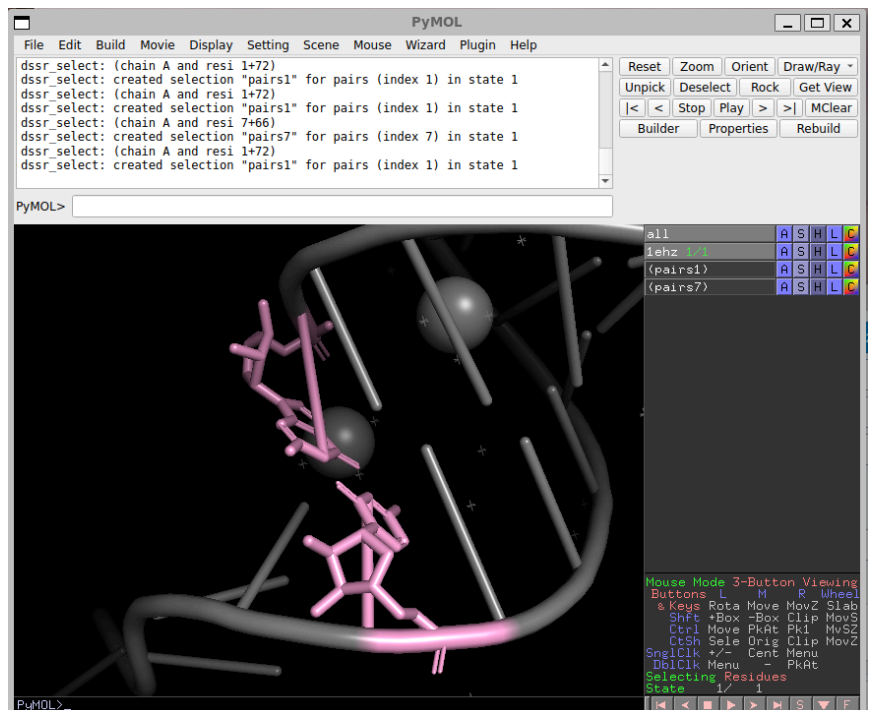
- The GUI basically has all of the parameters of the general `dssr_select` command built into the interface. By selecting the button labeled “Settings” in the top right corner of the GUI, more features are revealed to enhance visualization. Some of which are features directly from the `dssr_select` command (color, exe, selection, State, precolor, etc).



- One unique feature displayed here is the box labeled “Display sticks”. This will make selections appear in a stick-like format on PyMOL’s 3D viewer. For example, selecting the first index of a pair in the structure 1ehz shows the following in PyMOL’s 3D viewer:

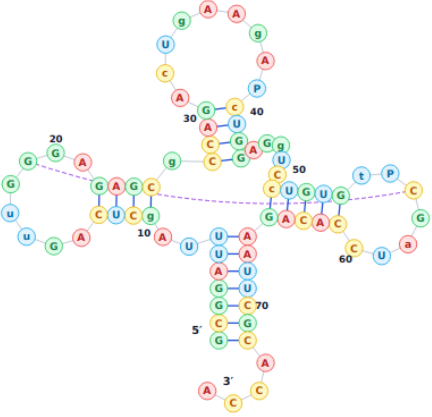
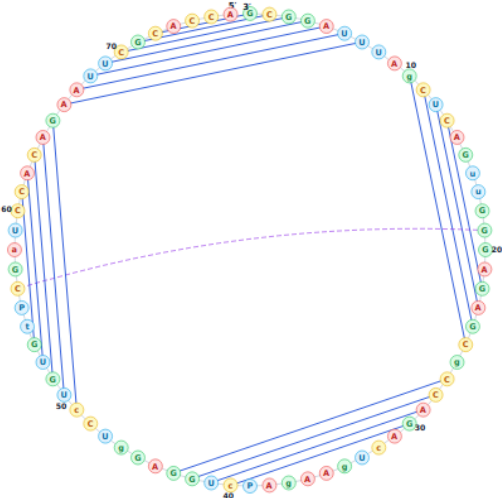


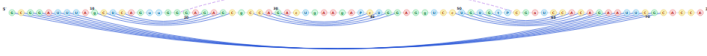
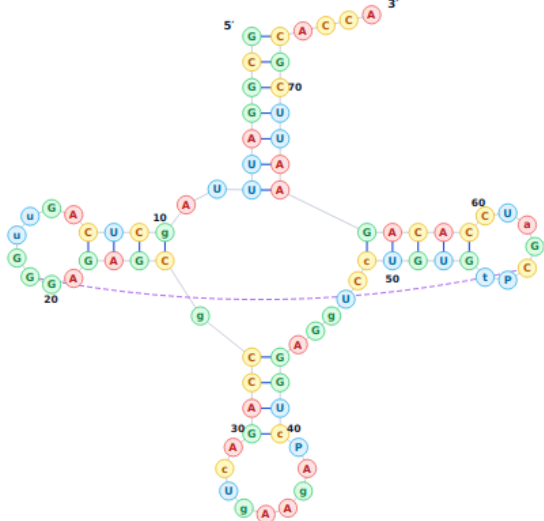
- This shows the selected pair as a set of sticks
- Another feature is the adjacent box to "Display Sticks" which is labeled as "Zoom to selection". This essentially means that when a feature is selected from the GUI, PyMOL's 3D viewer will zoom into the selection to show it up close. Using the same example of the first index of a pair in 1ehz, the image is up close:



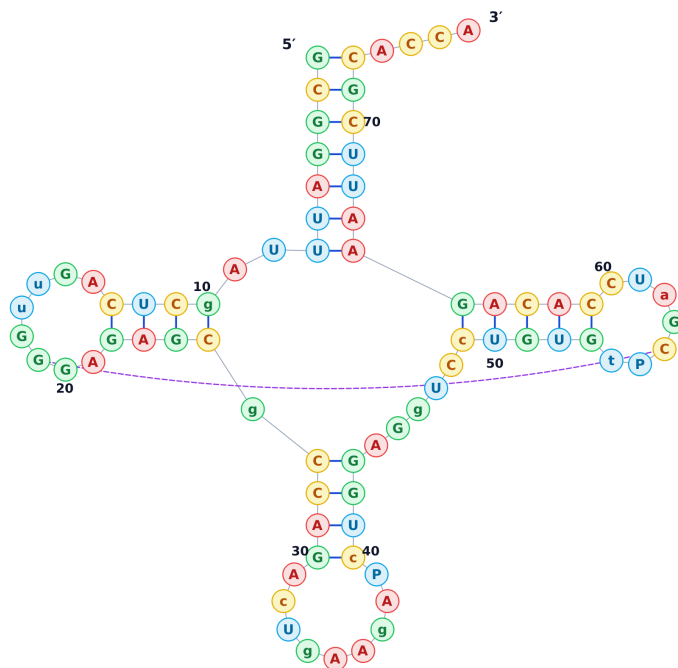
- The “Make Blocks” button effectively creates a cartoon block-like schematic that will be shown in PyMOL’s 3D viewer based on the user’s selection from the GUI. It is customizable in appearance through the parameters labeled as “Block style” which has a dropdown with multiple options and the “Depth” box which controls the thickness of created blocks. These features and what they do will be more understandable from reading the section on the `dssr_block` command
- Users can also manipulate the appearance of the 2D diagram as different types of diagrams can show different aspects. In the GUI, a dropdown box has “standard” on it as that is the default. By clicking on it, users can change the format of the 2D diagram to “circular”, “linear”, or “legacy radiate”. See the table for a comparison of each:

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Standard	
Circular	

Linear	
Legacy Radiate	

- Users are able to take the 2D diagrams of nucleic acid structures as well as their highlighted selections on those structures and export them. Using the “File” button, users get multiple options to save the image to their computer either as a .png file or as a .svg file.

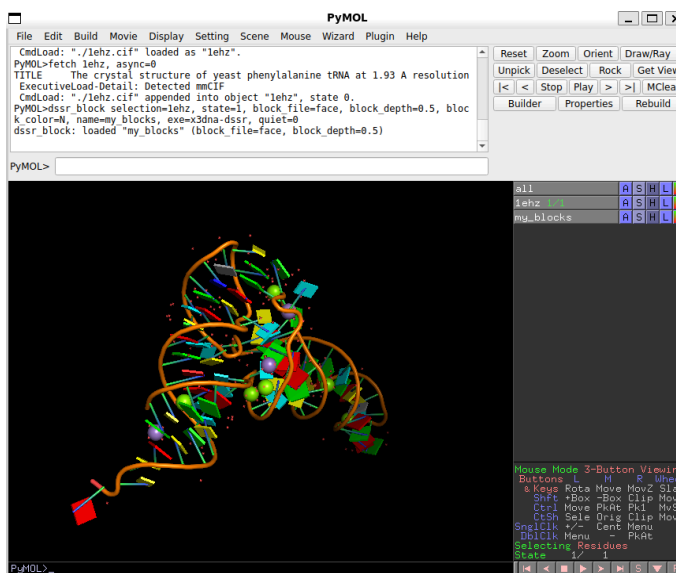


- The following schematic was obtained by exporting the image as a .png file from the GUI
- dssr_block: the command 'dssr_block' is used to create the highly informative and aesthetic cartoon block schematics of features within nucleic acid structures in PyMOL
 - The dssr_block command has the following parameters:

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Parameter	Default	Meaning
Selection	All	The atoms to make into block schematics
state	-1 (current)	The statue to use (almost never filled out)
block_file	face	The drawing style of the block
block_depth	0.5	The thickness of the blocks visually
block_color	(empty)	Can be used to override the color of nucleic acid bases
name	dssr_block## (auto)	Name for object
exe	x3dna-dssr	Path to DSSR executable

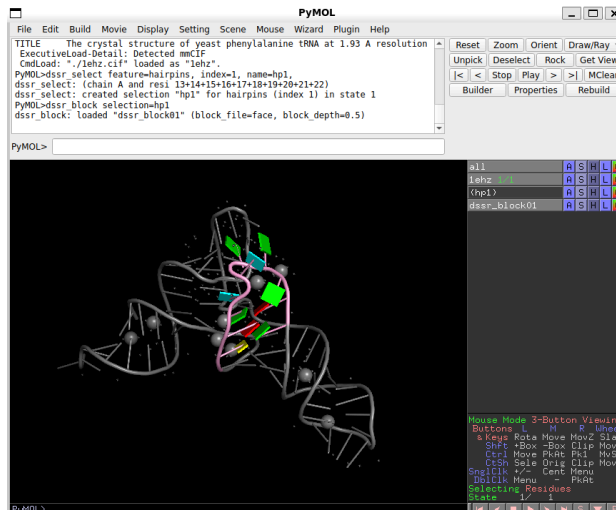
- To run dssr_block with all of its parameters filled out, run the following commands for making the entire 1ehz structure in block format:
 - fetch 1ehz
 - dssr_block selection=1ehz, state=1, block_file=face, block_depth=0.5, block_color=N, name=x, exe=x3dna-dssr, quiet=0



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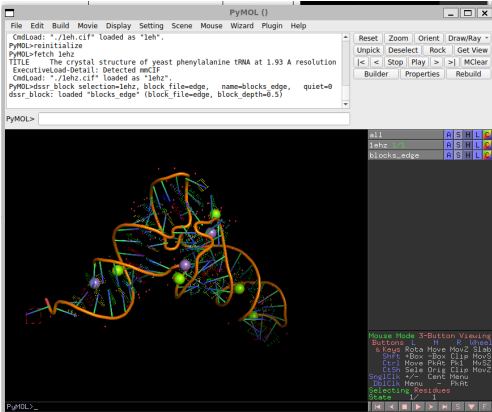
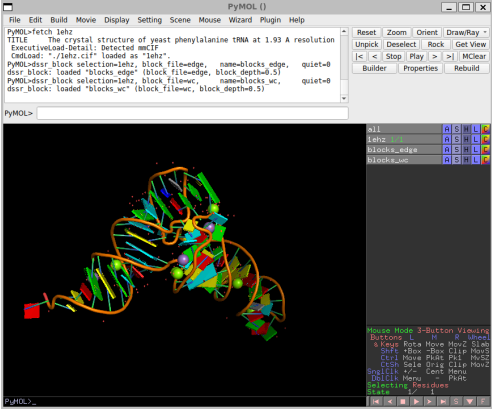
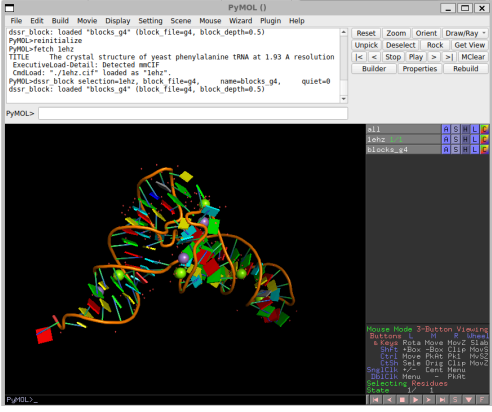
- dssr_block can also be used on a selection to make only that part of the nucleic acid structure in block format by creating and naming a selection first with dssr_select and running dssr_block on that selection.

- fetch 1ehz
- dssr_select feature=hairpins, index=1, name=hp1,
- dssr_block selection=hp1



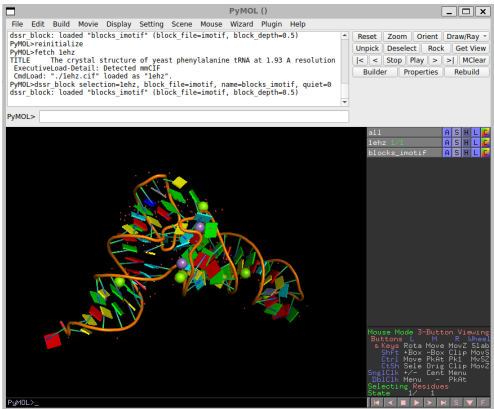
- dssr_block has multiple different block styles that all portray the cartoon block schematics differently as seen in the table below. First, fetch 1ehz in PyMOL. Then, use the table as a guide to creating the different type of block schematics:
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Command	Block Style	Image
dssr_block selection=1ehz block_file=face name=blocks_face quiet=0	face	

<pre>dssr_block selection=1ehz block_file=edge name=blocks_edge quiet=0</pre>	edge	
<pre>dssr_block selection=1ehz block_file=wc name=blocks_wc quiet=0</pre>	wc	
<pre>dssr_block selection=1ehz block_file=g4 name=blocks_g4 quiet=0</pre>	g4	

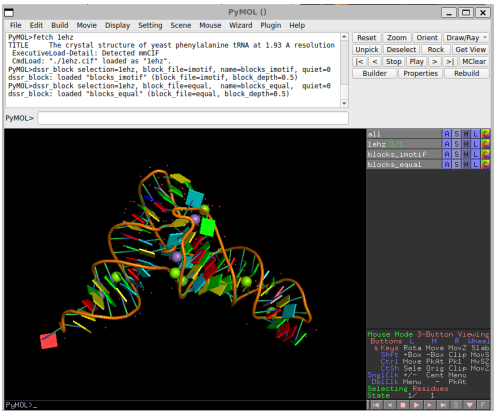
dssr_block
selection=1ehz
block_file=imotif
name=blocks_imotif
quiet=0

imotif



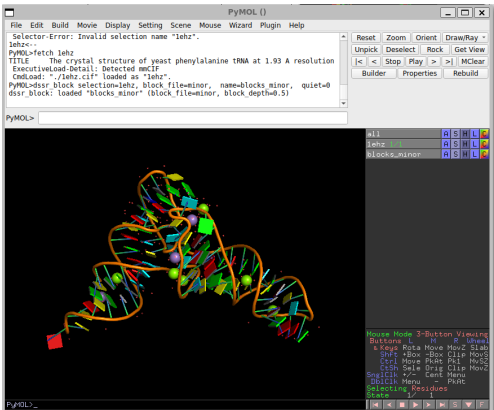
dssr_block
selection=1ehz
block_file=equal
name=blocks_equal
quiet=0

equal



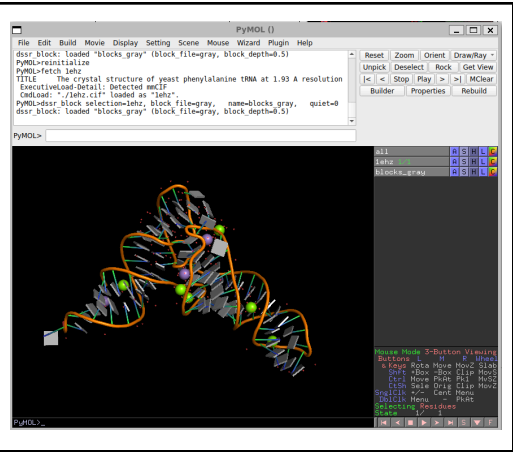
dssr_block
selection=1ehz
block_file=minor
name=blocks_minor
quiet=0

minor



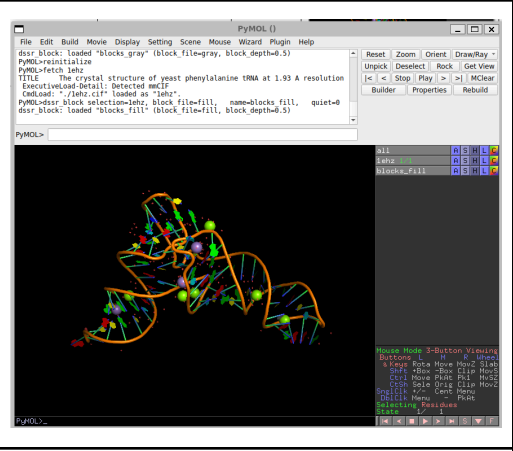
```
dssr_block  
selection=1ehz  
block_file=gray  
name=blocks_gray  
quiet=0
```

gray



```
dssr_block
selection=1ehz,
block_file=fill,
name=blocks_fill,
quiet=0
```

fill



```
dssr_block
selection=1ehz,
block_file=hbond,
name=blocks_hbond,
quiet=0
```

hbond

